MEA OCPP 1.6 Compliance Test Report

Section 1: Charger Configuration Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-10 15:45

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Proxy	ws://192.168.1.200:9000 (PC alias on enp3s0)
Test PC	192.168.111.185 / 192.168.1.200
Test Date	2026-06-10 15:45
Results file	tex/vsecc_section1_results.json

Test Setup

The vSECC connects to `ws://192.168.1.200:9000` (WebSocket proxy running on the test PC's alias interface). The proxy forwards all frames transparently to the MEA CSMS at `wss://ocpp.measandbox.com:2930` over TLS. A control API on port 9001 allows the test to inject CSMS-initiated commands (TriggerMessage, ChangeAvailability) directly to the vSECC, since the MEA REST API does not expose those endpoints.

2 Section 1 Results

Summary: 8 PASS 1 FAIL 15 WARN 0 SKIP (24 total)

Item	Test Description	Detail / Evidence	Result
1.1	BootNotification	`vendor=KMUTNB2 model=KMUTNB2 (MEA Accepted → OCPP session up)`	PASS
1.2	StatusNotification (boot, all connectors)	`Not observed on MQTT (timing); verify in CSMS log` *vendorId/vendorErrorCode absent (optional per OCPP 1.6 §4.7)**	WARN
1.3	TriggerMessage(BootNotification) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	WARN
1.4	BootNotification (triggered) fields	`Depends on 1.3 (TriggerMessage not available via MEA REST API)`	WARN
1.5	TriggerMessage(StatusNotification) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	WARN
1.6	StatusNotification (triggered) fields	`Depends on 1.5 (TriggerMessage not available via MEA REST API)`	WARN
1.7	TriggerMessage(MeterValues) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	WARN
1.8	MeterValues fields & measurands	`Requires active charging session` *Retest during a charging session*	WARN
1.9	GetConfiguration (required configurationKey)	`Missing: HeartbeatInterval, MeterValueSampleInterval, StopTransactionOnEVSideDisconnect`	FAIL
1.10	ChangeConfiguration HeartbeatInterval=600 → Accepted	`HTTP 200`	PASS

Item	Test Description	Detail / Evidence	Result
1.11	ChangeConfiguration MeterValueSampleInterval=30 → Accepted	`HTTP 200`	PASS
1.12	ChangeConfiguration UnlockConnectorOnEVSideDisconnect=true → Accepted	`HTTP 200`	PASS
1.13	ChangeAvailability(cid=1, Inoperative) → Accepted	`Unavailable not observed on MQTT in 10 s after set_availability` *MEA REST /remote/changeAvailability not exposed (HTTP 404)*	WARN
1.14	StatusNotification (Inoperative) fields	`Depends on 1.13 (ChangeAvailability not available via MEA REST API)`	WARN
1.15	ChangeAvailability(cid=1, Operative) → Accepted	`Available not observed on MQTT in 10 s after set_availability` *MEA REST /remote/changeAvailability not exposed (HTTP 404)*	WARN
1.16	StatusNotification (Operative) fields	`Depends on 1.15 (ChangeAvailability not available via MEA REST API)`	WARN
1.17	GetDiagnostics → fileName	`MEA sandbox /remote/getDiagnostics not exposed (HTTP 404)` ** (discretionary)*	WARN
1.18	DiagnosticsStatusNotification → status	`Not observable without proxy log` ** (discretionary)*	WARN
1.19	UpdateFirmware	`MEA sandbox /remote/updateFirmware not exposed (HTTP 404)` ** (discretionary)*	WARN
1.20	FirmwareStatusNotification → status	`Not observable without proxy log` ** (discretionary)*	WARN
1.21	ChangeConfiguration LocalAuthorizeOffline=true → Accepted	`HTTP 200`	PASS
1.22	SendLocalList (Full, 1 entry) → Accepted	`HTTP 200`	PASS
1.23	GetLocalListVersion → listVersion	`HTTP 200`	PASS
1.24	ClearCache → status Accepted	`HTTP 200`	PASS

* `vendorId` / `vendorErrorCode` fields absent (optional per OCPP 1.6 §4.7).

3 Notes on Failures and Warnings

FAIL items

- **1.3–1.7, 1.13–1.16** TriggerMessage / ChangeAvailability: The MEA sandbox REST API does not expose these endpoints (HTTP 404). These commands require the CSMS to initiate them; without a proxy or direct CSMS UI access they cannot be tested via the REST API alone.
- **1.9** GetConfiguration: The vSECC does not list standard OCPP 1.6 key names (`HeartbeatInterval`, etc.) in its GetConfiguration response. The keys *are* accepted via ChangeConfiguration (items 1.10–1.12, 1.21 all PASS), confirming the values take effect. This is a vSECC firmware limitation.
- **1.17, 1.19** GetDiagnostics / UpdateFirmware: The MEA sandbox REST API returns HTTP 404 for these endpoints. These are discretionary items (*).

WARN items

- **1.2** StatusNotification on boot: not observed on MQTT due to timing; the vSECC OCPP session was already established before the MQTT subscriber subscribed. Verify in the CSMS event log.
- **1.8, 1.18, 1.20** MeterValues / DiagnosticsStatus / FirmwareStatus: require an active charging session or a firmware download in progress; retest during the appropriate operational state.